CS341 Software Engineering: Task Tracker

Project 1 — Plain English Project Presentation

1 November 2026

Problem and Scope

- Small teams need to assign and track tasks.
- Single-machine Java desktop application.
- Multiple accounts use one local SQLite file.
- No real-time online synchronization.

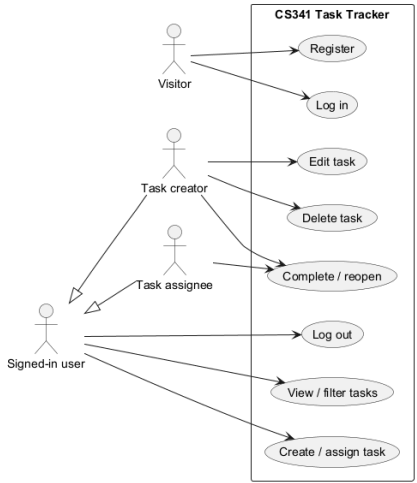
Assignment Requirements

- Register, log in and log out.
- Create, assign, edit, delete and complete tasks.
- View created or assigned tasks.
- Persist data with SQLite through JDBC.
- Use Git, JUnit 5 and documented system design.

Additional Features

- Three priority levels: LOW, MEDIUM, HIGH.
- Optional due dates in YYYY-MM-DD format.
- Open / completed / high-priority filters.
- Sorting by task table columns.

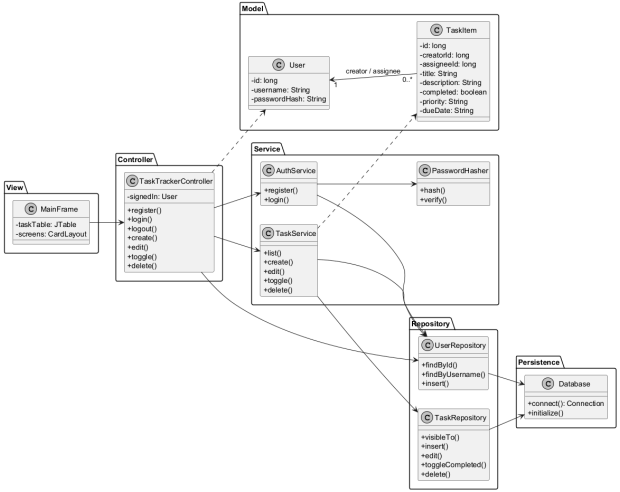
Use Cases


CS341 Task Tracker
Visitor
Register
Log in
Task creator
Edit task
Delete task
Task assignee
Complete / reopen
Signed-in user
Log out
View / filter tasks
Create / assign task

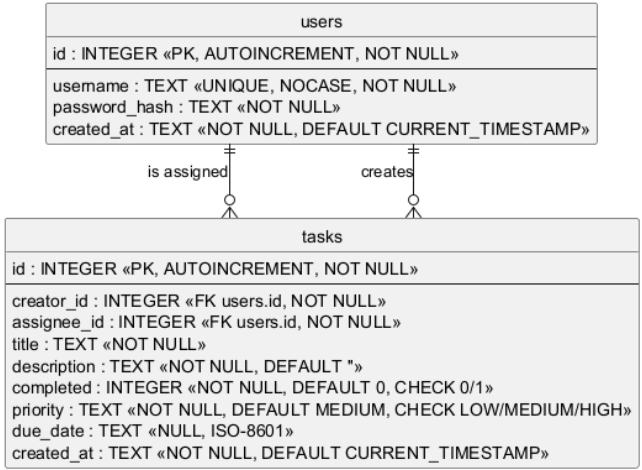
Architecture

- Swing View: MainFrame.
- Controller: TaskTrackerController.
- Business rules: AuthService and TaskService.
- Storage access: repositories and Database.

Class Diagram



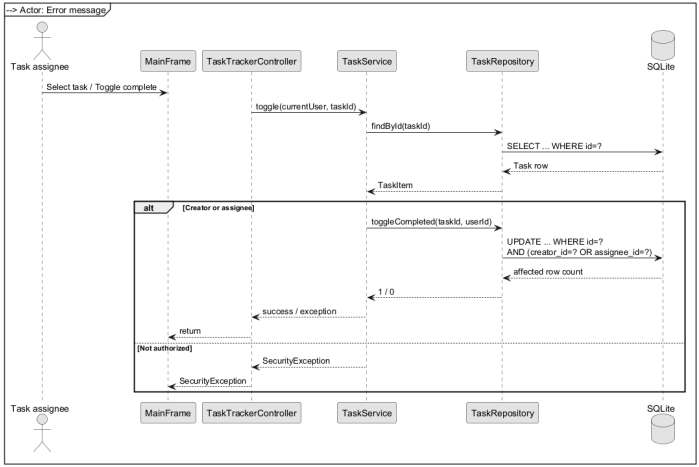
SQLite Data Model



Authorization

- Creator: view, edit, delete and toggle completion.
- Assignee: view and toggle completion.
- Unrelated user: no access.
- SQL writes have user-ID conditions as a second guard.

Task Completion Sequence



Implementation Details

- Java 11+, Swing, JDBC and SQLite.
- Parameterized SQL and database foreign keys.
- Salted PBKDF2 password hashes.
- Automatic creation of users and tasks tables.

Automated Tests

- Maven/Surefire: 17 JUnit 5 tests passed; 0 failures, errors or skips.
- Packaged JAR SQLite JDBC create/insert/read smoke test passed.
- Independent Python SQLite checks: 10/10 passed.
- Final ZIP extracted and passed a clean Maven build.

Manual Demo Script

1. Register Alice and Bob.
2. Alice assigns a task to Bob.
3. Bob completes the task but cannot edit/delete.
4. Alice changes priority, reassigns or deletes the task.
5. Restart: verify persistence and task filters.

Design Trade-offs

- Local database: low setup cost, offline use.
- No multi-device collaboration or backup automation.
- Permission policy is a stated design choice.
- Lean MVC-oriented packages for small-coursework scope.

Open Verification Gates

- Run AT-01–AT-08 on a real desktop and capture genuine GUI evidence.
- Provide authentic 3–4 member contributions and Git history.
- Publish only with team approval and record the real remote CI result.
- Deliver the live demo with all actual team members.

Conclusion

- Local Maven/JUnit/JDBC, packaged-JAR SQLite and extracted ZIP builds passed.
- Seven PlantUML sources were rendered; six English PDFs were rebuilt.
- GUI acceptance and real team/GitHub submission remain open.

Questions?